malodor. Periodontal disease, caries, acute forms of gingivitis, poorly fitting dentures, oral abscess, and tongue coating are common causes. Treatment includes correcting poor hygiene, treating infection, and tongue brushing. Hyposalivation can produce and exacerbate halitosis. Pockets of decay in the tonsillar crypts, esophageal diverticulum, esophageal stasis (e.g., achalasia, stricture), sinusitis, and lung abscess account for some instances. A few systemic diseases produce distinctive odors: renal failure and *Helicobacter pylori* gastritis (ammoniacal), hepatic (fishy), and ketoacidosis (fruity). If a patient presents because of concern about halitosis but no odor is detectable, then pseudohalitosis or halitophobia must be considered.

Aging and Oral Health While tooth loss and dental disease are not normal consequences of aging, a complex array of structural and functional changes that occur with age can affect oral health. Subtle changes in tooth structure (e.g., diminished pulp space and volume, sclerosis of dentinal tubules, and altered proportions of nerve and vascular pulp content) result in the elimination or diminution of pain sensitivity and a reduction in the reparative capacity of the teeth. In addition, age-associated fatty replacement of salivary acini may reduce physiologic reserve, thus increasing the risk of hyposalivation. In healthy older adults, there is minimal, if any, reduction in salivary flow.

Poor oral hygiene often results when general health fails or when patients lose manual dexterity and upper-extremity flexibility. This situation is particularly common among frail older adults and nursing home residents, and regular oral cleaning and dental care reduce the incidence of pneumonia, oral disease, the mortality risk in this population. Other risks for dental decay include limited lifetime fluoride exposure. Without assiduous care, decay can become quite advanced yet remain asymptomatic. Consequently, much of a tooth—or the entire tooth—can be destroyed before the patient is aware of the process.

Periodontal disease, a leading cause of tooth loss, is indicated by loss of alveolar bone height. More than 90% of the U.S. population has some degree of periodontal disease by age 50. Healthy adults who have not had significant alveolar bone loss by the sixth decade of life do not typically experience significant worsening with advancing age.

Complete edentulousness in the United States is becoming less common in the elderly and is increasingly restricted to impoverished populations. When it is present, speech, mastication, and facial contours are dramatically affected. Edentulousness may also exacerbate obstructive sleep apnea, particularly in asymptomatic individuals who wear dentures. Dentures can improve verbal articulation and restore diminished facial contours. Mastication can also be restored; however, patients expecting dentures to facilitate oral intake are often disappointed. Accommodation to dentures requires a period of adjustment. Pain can result from friction or traumatic lesions produced by loose dentures. Poor fit and poor oral hygiene may permit the development of candidiasis. This fungal infection may be either asymptomatic or painful and is suggested by erythematous smooth or granular tissue conforming to an area covered by the appliance. Individuals with dentures and no natural teeth need regular (annual) professional oral examinations.

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Section 5 Alterations in Circulatory and Respiratory Functions

39 Dyspnea

Rebecca M. Baron

DYSPNEA

DEFINITION

The American Thoracic Society consensus statement defines *dyspnea* as a "subjective experience of breathing discomfort that consists of qualitatively distinct sensations that vary in intensity. The experience derives from interactions among multiple physiological, psychological, social, and environmental factors and may induce secondary physiological and behavioral responses." Dyspnea, a symptom, can be perceived only by the person experiencing it and, therefore, must be self-reported. In contrast, signs of increased work of breathing, such as tachypnea, accessory muscle use, and intercostal retraction, can be measured and reported by clinicians.

EPIDEMIOLOGY

Dyspnea is common. It has been reported that up to one-half of inpatients and one-quarter of ambulatory patients experience dyspnea, with a prevalence of 9–13% in the community that increases to as high as 37% for adults aged ≥70 years. Dyspnea is a frequent cause of emergency room visits, accounting for as many as 3–4 million visits per year. Furthermore, it is increasingly appreciated that the degree of dyspnea may better predict outcomes in chronic obstructive pulmonary disease (COPD) than does the forced expiratory volume in 1 s (FEV_1), and formal measures of dyspnea have been incorporated into the Global Initiative for Chronic Obstructive Lung Disease (GOLD) COPD severity assessment guidelines. Dyspnea may also predict outcomes in other chronic heart and lung diseases as well. Dyspnea can arise from a diverse array of pulmonary, cardiac, and neurologic underlying causes, and elucidation of particular symptoms may point toward a specific etiology and/or mechanism driving dyspnea (although additional diagnostic testing is often required, as will be further discussed below). There has been an increased focus on dyspnea given its increased incidence in the setting of the SARS-CoV-2 pandemic, and post-COVID persistent symptoms in many patients (Chap. 205).

MECHANISMS UNDERLYING DYSPNEA

The mechanisms underlying dyspnea are complex, as it can arise from different contributory respiratory sensations. Although a large body of research has increased our understanding of mechanisms underlying particular respiratory sensations such as "chest tightness" or "air hunger," it is likely that a given disease state might produce the sensation of dyspnea via more than one underlying mechanism. Dyspnea can arise from a variety of pathways, including generation of *afferent* signals from the respiratory system to the central nervous system (CNS), *efferent* signals from the CNS to the respiratory muscles, and particularly when there is a mismatch in the integrative signaling between these two pathways, termed *efferent-reafferent mismatch* (Fig. 39-1).

Afferent signals trigger the CNS (brainstem and/or cortex) and include primarily: (1) peripheral chemoreceptors in the carotid body and aortic arch and central chemoreceptors in the medulla that are activated by hypoxemia, hypercapnia, or acidemia, and might produce a sense of "air hunger"; and (2) mechanoreceptors in the upper airways, lungs (including stretch receptors, irritant receptors, and J receptors), and chest wall (including muscle spindles as stretch receptors and tendon organs that monitor force generation) that are activated in the setting of an increased workload from a disease state producing an increase in airway resistance that may be associated with symptoms of chest tightness (e.g., asthma or COPD) or decreased lung or chest wall

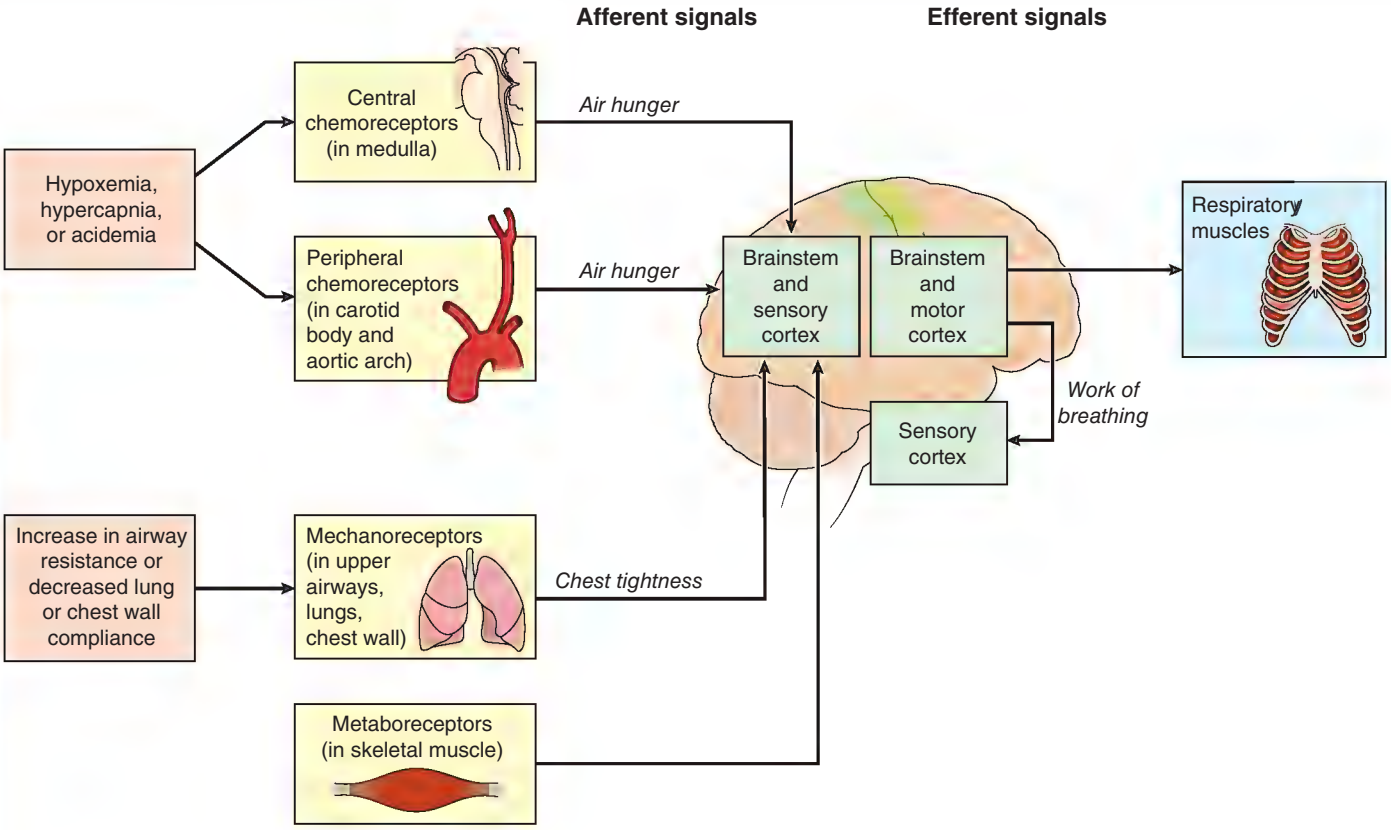

FIGURE 39-1 Potential signaling pathways underlying the sensation of dyspnea. Dyspnea is a complex sensation that can arise from a variety of physiologic stimuli (examples in pink boxes include hypoxemia, hypercapnia, acidemia, increase in airway resistance, and decrease in lung and/or chest wall compliance). Physiologic signals are mediated via relevant receptors (examples in yellow boxes include central chemoreceptors in the medulla; peripheral chemoreceptors in the carotid body and aortic arch; mechanoreceptors in upper airways, lungs, and chest wall; and metaboreceptors in skeletal muscle) that send afferent signals to the brainstem and sensory cortex. Efferent signals are sent from the brainstem and motor cortex to the respiratory muscles and sensory cortex. Potential symptoms associated with dyspnea that may be generated by these signals are indicated in italics.

compliance (e.g., pulmonary fibrosis). Other afferent signals that trigger dyspnea within the respiratory system can arise from pulmonary vascular receptor responses to changes in pulmonary artery pressure and skeletal muscle (termed metaboreceptors) that are believed to sense changes in the biochemical environment.

Efferent signals are sent from the CNS (motor cortex and brainstem) to the respiratory muscles and are also transmitted by corollary discharge to the sensory cortex; they are believed to underlie sensations of respiratory effort (or "work of breathing") and perhaps contribute to sensations of "air hunger," especially in response to an increased ventilatory load in a disease state such as COPD. In addition, fear or anxiety may heighten the sense of dyspnea by exacerbating the underlying physiologic disturbance in response to an increased respiratory rate or disordered breathing pattern.

ASSESSING DYSPNEA

While it is well appreciated that dyspnea is a difficult quality to measure reliably owing to multiple relevant possible domains that can be measured (e.g., sensory-perceptual experience, affective distress, and symptom impact or burden) and that there are no uniformly agreed upon tools for dyspnea assessment, consensus opinion holds that dyspnea should be formally assessed in a context most relevant and beneficial for patient management and, furthermore, that the specific domains being measured are adequately described. There are a number of emerging tools that have been developed for formal dyspnea assessment. As an example, the GOLD criteria advocate use of a dyspnea assessment tool such as the Modified Medical Research Council Dyspnea Scale (Table 39-1) to assess symptom/impact burden in COPD.

DIFFERENTIAL DIAGNOSIS

This chapter focuses largely on chronic dyspnea, which is defined as symptoms lasting longer than 1 month and can arise from a broad array of different underlying conditions, most commonly attributable to pulmonary or cardiac conditions that account for as many as 85% of the underlying causes of dyspnea. However, as many as one-third of patients may have multifactorial reasons underlying dyspnea, with an increasing number of individuals suffering from dyspnea as part of a post-COVID syndrome (Chap. 205). Examples of a wide array of conditions that underlie dyspnea with possible mechanisms underlying the presenting symptoms are described in Table 39-2.

Respiratory system causes include diseases of the airways (e.g., asthma and COPD), diseases of the parenchyma (more commonly, interstitial lung diseases are seen in the setting of chronic dyspnea,

TABLE 39-1 An Example of a Clinical Method for Rating Dyspnea: The Modified Medical Research Council Dyspnea Scale[a]

GRADE OF DYSPNEA	DESCRIPTION
0	Not troubled by breathlessness, except with strenuous exercise
1	Shortness of breath walking on level ground or with walking up a slight hill
2	Walks slower than people of similar age on level ground due to breathlessness, or has to stop to rest when walking at own pace on level ground
3	Stops to rest after walking 100 m or after walking a few minutes on level ground
4	Too breathless to leave the house, or breathless with activities of daily living (e.g., dressing/undressing)

[a]This scale has been integrated into the Global Initiative for Chronic Obstructive Lung Disease (GOLD) guidelines clinical classification scheme.

Source: Reproduced with permission from DA Mahler, CK Wells: Evaluation of clinical methods for rating dyspnea. Chest 93:580, 1988.

TABLE 39-2 Differential Diagnosis of Disease Processes Underlying Dyspnea

SYSTEM	TYPE OF PROCESS	EXAMPLE OF DISEASE PROCESS	POSSIBLE PRESENTING DYSPNEA SYMPTOMS	POSSIBLE PHYSICAL FINDINGS	POSSIBLE MECHANISMS UNDERLYING DYSPNEA	INITIAL DIAGNOSTIC STUDIES (AND POSSIBLE FINDINGS)
Pulmonary	Airways disease	Asthma, COPD, upper airway obstruction	Chest tightness, tachypnea, increased WOB, air hunger, inability to get a deep breath	Wheezing, accessory muscle use, exertional hypoxemia (especially with COPD)	Increased WOB, hypoxemia, hypercapnia, stimulation of pulmonary receptors	Peak flow (reduced); spirometry (OVD); CXR (hyperinflation; loss of lung parenchyma in COPD), chest CT and airway examination for upper airway obstruction
	Parenchymal disease	Interstitial lung disease[a]	Air hunger, inability to get a deep breath	Dry end-inspiratory crackles, clubbing, exertional hypoxemia	Increased WOB, increased respiratory drive, hypoxemia, hypercapnia, stimulation of pulmonary receptors	Spirometry and lung volumes (RVD); CXR and chest CT (interstitial lung disease)
	Chest wall disease	Kyphoscoliosis, neuromuscular (NM) weakness	Increased WOB, inability to get a deep breath	Decreased diaphragm excursion; atelectasis	Increased WOB; stimulation of pulmonary receptors (if atelectasis is present)	Spirometry and lung volumes (RVD); MIP and MEPs (reduced in NM weakness)
Pulmonary and cardiac	Pulmonary vasculature	Pulmonary hypertension	Tachypnea	Elevated right heart pressures, exertional hypoxemia	Increased respiratory drive, hypoxemia, stimulation of vascular receptors	Diffusion capacity (reduced); ECG; ECHO, RHC (to evaluate pulmonary artery pressures)[b]
Cardiac	Left heart failure Pericardial disease	Coronary artery disease, cardiomyopathy[c] Constrictive pericarditis; cardiac tamponade	Chest tightness, air hunger	Elevated left heart pressures; wet crackles on lung examination; pulsus paradoxus (pericardial disease)	Increased WOB and drive, hypoxemia, stimulation of vascular and pulmonary receptors[d]	Consider BNP testing, especially in the acute setting; ECG, ECHO, may need stress testing and/or LHC
Other	Variable	Anemia Deconditioning Psychological Metabolic disturbances Gastrointestinal (e.g., gastroesophageal reflux disease [GERD], aspiration pneumonitis) Post-COVID syndrome	Exertional breathlessness Poor fitness Anxiety	Variable	Metaboreceptors (anemia, poor fitness); chemoreceptors (anaerobic metabolism from poor fitness); some subjects may have increased sensitivity to hypercapnia	Hematocrit for anemia; laboratory studies (e.g., metabolic panel, thyroid hormone testing for metabolic disturbances); consider upper gastrointestinal endoscopy and/or esophageal pH probe testing for GERD and concerns for aspiration; consider referral to post-COVID care center for persistent symptoms after COVID infection; exclude other causes

[a]Differential diagnosis of interstitial lung disease includes idiopathic pulmonary fibrosis, collagen vascular disease, drug- or occupation-induced pneumonitis, lymphangitic spread of malignancy; processes that are more alveolar rather than interstitial in nature can also less commonly contribute to parenchymal lung disease underlying chronic dyspnea and include entities such as hypersensitivity pneumonitis, bronchiolitis obliterans organizing pneumonia, etc. [b]Would additionally consider these patients for CT angiography to evaluate for presence of thromboemboli and ventilation/perfusion scanning to evaluate for the presence of chronic thromboembolic disease. [c]Diastolic dysfunction in the setting of a stiff left ventricle is often seen and contributes significantly to insidious dyspnea that can be difficult to treat. [d]May stimulate metaboreceptors if cardiac output is sufficiently reduced to result in a lactic acidosis.

Abbreviations: BNP, brain natriuretic peptide; COPD, chronic obstructive pulmonary disease; COVID, coronavirus disease; CT, computed tomography; CXR, chest x-ray; ECG, electrocardiogram; ECHO, echocardiogram; GERD, gastroesophageal reflux disease; LHC, left heart catheterization; MIP/MEP, maximal inspiratory and maximal expiratory pressures (obtained in the pulmonary function testing laboratory); OVD, obstructive ventilatory defect; RVD, restrictive ventilatory defect; RHC, right heart catheterization; WOB, work of breathing.

but alveolar filling processes, such as hypersensitivity pneumonitis or bronchiolitis obliterans organizing pneumonia [BOOP], can also present with similar symptoms), diseases affecting the chest wall (e.g., bony abnormalities such as kyphoscoliosis, or neuromuscular weakness conditions such as amyotrophic lateral sclerosis), and diseases affecting the pulmonary vasculature (e.g., pulmonary hypertension that can arise from a variety of underlying causes, or chronic thromboembolic disease). Diseases affecting the cardiovascular system that can present with dyspnea include processes affecting left heart function, such as coronary artery disease and cardiomyopathy, as well as disease processes affecting the pericardium, including constrictive pericarditis and cardiac tamponade. Other conditions underlying dyspnea that might not directly emanate from the pulmonary or cardiovascular systems include anemia (thereby potentially affecting oxygen-carrying capacity), deconditioning, and psychological processes such as anxiety. Distinguishing among the myriad of underlying processes that might present with dyspnea can be challenging. A graded approach that begins with a history and physical examination, followed by selected laboratory testing that might then advance to additional diagnostics and potentially subspecialty referral, may help elucidate the underlying cause of dyspnea. However, a substantial proportion of patients may have persistent dyspnea despite treatment for an underlying process or may not have a specific underlying process identified that is driving the dyspnea.

APPROACH TO THE PATIENT

Dyspnea (See Fig. 39-2)

OVERALL

For patients with a known prior pulmonary, cardiac, or neuromuscular condition and worsening dyspnea, the initial focus of the evaluation will usually address determining whether the known

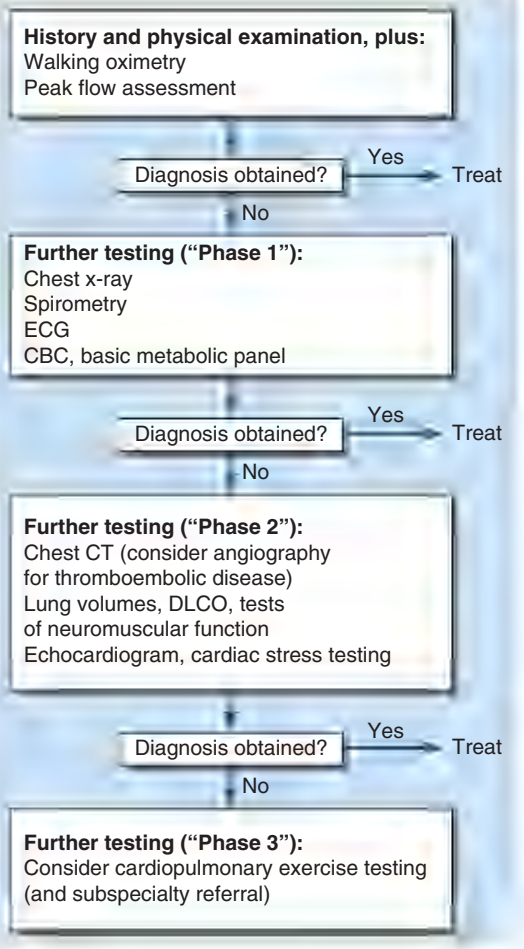

FIGURE 39-2 Possible algorithm for the evaluation of the patient with dyspnea. As described in the text, the approach should begin with a detailed history and physical examination, followed by progressive testing and ultimately more invasive testing and subspecialty referral as is indicated to determine the underlying cause of dyspnea. CBC, complete blood count; DLCO, diffusing capacity of the lungs for carbon monoxide; ECG, electrocardiogram. *(Adapted from NG Karnani et al: Am Fam Physician 71:1529, 2005.)*

condition has progressed or whether a new process has developed that is causing dyspnea. For patients without a prior known potential cause of dyspnea, the initial evaluation will focus on determining an underlying etiology. Determining the underlying cause, if possible, is extremely important, as the treatment may vary dramatically based on the predisposing condition. An initial history and physical examination remain fundamental to the evaluation followed by initial diagnostic testing as indicated that might prompt subspecialty referral (e.g., pulmonary, cardiology, neurology, sleep, and/or specialized dyspnea clinic) if the cause of dyspnea remains elusive (Fig. 39-2). As many as two-thirds of patients will require diagnostic testing beyond the initial clinical presentation.

HISTORY

The patient should be asked to describe in their own words what the discomfort feels like as well as the effect of position, infections, and environmental stimuli on the dyspnea, as descriptors may be helpful in pointing toward an etiology. For example, symptoms of chest tightness might suggest the possibility of bronchoconstriction, and the sensation of inability to take a deep breath may correlate with dynamic hyperinflation from COPD. Orthopnea is a common indicator of congestive heart failure (CHF), mechanical impairment of the diaphragm associated with obesity, or asthma triggered by esophageal reflux. Nocturnal dyspnea suggests CHF or asthma. Acute, intermittent episodes of dyspnea are more likely to reflect episodes of myocardial ischemia, bronchospasm, or pulmonary embolism, whereas chronic persistent dyspnea is more typical of COPD, interstitial lung disease, and chronic thromboembolic disease. Information on risk factors for drug-induced or occupational lung disease and for coronary artery disease should be elicited. Left atrial myxoma or hepatopulmonary syndrome should be considered when the patient complains of *platypnea*—i.e., dyspnea in the upright position with relief in the supine position.

PHYSICAL EXAMINATION

Initial vital signs might be helpful in pointing toward an underlying etiology in the context of the remainder of the evaluation. For example, the presence of fever might point toward an underlying infectious or inflammatory process; the presence of hypertension in the setting of a heart failure might point toward diastolic dysfunction; the presence of tachycardia might be associated with many different underlying processes including fever, cardiac dysfunction, and deconditioning; and the presence of resting hypoxemia suggests processes involving hypercapnia, ventilation-perfusion mismatch, shunt, or impairment in diffusion capacity might be involved. An exertional oxygen saturation should also be obtained as described below. The physical examination should begin during the interview of the patient. Inability of the patient to speak in full sentences before stopping to get a deep breath suggests a condition that leads to stimulation of the controller or impairment of the ventilatory pump with reduced vital capacity. Evidence of increased work of breathing (supraclavicular retractions; use of accessory muscles of ventilation; and the tripod position, characterized by sitting with the hands braced on the knees) is indicative of increased airway resistance or stiffness of the lungs and the chest wall. When measuring the vital signs, the physician should accurately assess the respiratory rate and measure the pulsus paradoxus **(Chap. 281)**; if the systolic pressure decreases by >10 mmHg on inspiration, the presence of COPD, acute asthma, or pericardial disease should be considered. During the general examination, signs of anemia (pale conjunctivae), cyanosis, and cirrhosis (spider angiomata, gynecomastia) should be sought. Examination of the chest should focus on symmetry of movement; percussion (dullness is indicative of pleural effusion; hyperresonance is a sign of pneumothorax and emphysema); and auscultation (wheezes, rhonchi, prolonged expiratory phase, and diminished breath sounds are clues to disorders of the airways; rales suggest interstitial edema or fibrosis). The cardiac examination should focus on signs of elevated right heart pressures (jugular venous distention, edema, accentuated pulmonic component to the second heart sound); left ventricular dysfunction (S3 and S4 gallops); and valvular disease (murmurs). When examining the abdomen with the patient in the supine position, the physician should note whether there is paradoxical movement of the abdomen as well as the presence of increased respiratory distress in the supine position: inward motion during inspiration is a sign of diaphragmatic weakness, and rounding of the abdomen during exhalation is suggestive of pulmonary edema. Clubbing of the digits may be an indication of interstitial pulmonary fibrosis or bronchiectasis, and joint swelling or deformation as well as changes consistent with Raynaud's disease may be indicative of a collagen-vascular process that can be associated with pulmonary disease.

Patients should be asked to walk under observation with oximetry in order to reproduce the symptoms. The patient should be examined during and at the end of exercise for new findings that were not present at rest (e.g., presence of wheezing) and for changes in oxygen saturation.

CHEST IMAGING

After the history elicitation and the physical examination, a chest radiograph should be obtained if the diagnosis remains elusive. The lung volumes should be assessed: hyperinflation is consistent with obstructive lung disease, whereas low lung volumes suggest interstitial edema or fibrosis, diaphragmatic dysfunction, or

impaired chest wall motion. The pulmonary parenchyma should be examined for evidence of interstitial disease, infiltrates, and emphysema. Prominent pulmonary vasculature in the upper zones indicates pulmonary venous hypertension, while enlarged central pulmonary arteries may suggest pulmonary arterial hypertension. An enlarged cardiac silhouette can point toward dilated cardiomyopathy or valvular disease. Bilateral pleural effusions are typical of CHF and some forms of collagen-vascular disease. Unilateral effusions raise the specter of carcinoma and pulmonary embolism but may also occur in heart failure or in the case of a parapneumonic effusion. CT of the chest is generally reserved for further evaluation of the lung parenchyma (e.g., interstitial lung disease) and possible pulmonary embolism (with CT angiography) if diagnostic uncertainty remains.

LABORATORY STUDIES

Initial laboratory testing should include a hematocrit to exclude occult anemia as an underlying cause of reduced oxygen-carrying capacity contributing to dyspnea, and a basic metabolic panel may be helpful to exclude a significant underlying metabolic acidosis (and conversely, an elevated bicarbonate might point toward the possibility of carbon dioxide retention that might be seen in chronic respiratory failure—in such a setting, an arterial blood gas may provide useful additional information). Additional laboratory studies should include electrocardiography to seek evidence of ventricular hypertrophy and prior myocardial infarction, and spirometry, which can be diagnostic of the presence of an obstructive ventilatory defect and suggest the possibility of a restrictive ventilatory defect (that then might prompt additional pulmonary function laboratory testing, including lung volumes, diffusion capacity, and possible tests of neuromuscular function). Echocardiography is indicated when systolic dysfunction, pulmonary hypertension, or valvular heart disease is suspected. Bronchoprovocation testing and/or home peak-flow monitoring may be useful in patients with intermittent symptoms suggestive of asthma who have a normal physical examination and spirometry; up to one-third of patients with the clinical diagnosis of asthma do not have reactive airways disease when formally tested. Measurement of brain natriuretic peptide levels in serum is increasingly used to assess for CHF in patients presenting with acute dyspnea but may be elevated in the presence of right ventricular strain as well.

DISTINGUISHING CARDIOVASCULAR FROM RESPIRATORY SYSTEM DYSPNEA

If a patient has evidence of both pulmonary and cardiac disease that is not responsive to treatment or it remains unclear what factors are primarily driving the dyspnea, a cardiopulmonary exercise test (CPET) can be conducted to determine which system is responsible for the exercise limitation. CPET includes incremental symptom-limited exercise (cycling or treadmill) with measurements of ventilation and pulmonary gas exchange and, in some cases, includes noninvasive and invasive measures of pulmonary vascular pressures and cardiac output. If, at peak exercise, the patient achieves predicted maximal ventilation, demonstrates an increase in dead space or hypoxemia, or develops bronchospasm, the respiratory system may be the cause of the problem. Alternatively, if the heart rate is >85% of the predicted maximum, if the anaerobic threshold occurs early, if the blood pressure becomes excessively high or decreases during exercise, if the O_2 pulse (O_2 consumption/heart rate, an indicator of stroke volume) falls, or if there are ischemic changes on the electrocardiogram, an abnormality of the cardiovascular system is likely the explanation for the breathing discomfort. Additionally, a CPET may also help point toward a peripheral extraction deficit or metabolic/neuromuscular disease as potential underlying processes driving dyspnea.

TREATMENT

Dyspnea

The first goal is to correct the underlying condition(s) driving dyspnea and address potentially reversible causes with appropriate treatment for the particular condition. Multiple different interventions may be necessary, given that dyspnea often arises from multifactorial causes. If relief of dyspnea with treatment of the underlying condition(s) is not fully possible, an effort is made to lessen the intensity of the symptom and its effect on the patient's quality of life. More recent work at the consensus conference level has sought to define an identifiable entity of persistent dyspnea in order to develop an approach to improving efforts to address symptom management for this condition. In 2017, an international group of experts defined "chronic breathlessness syndrome" as "the experience of breathlessness that persists despite optimal treatment of the underlying pathophysiology and results in disability for the patient." Despite an increased understanding of the mechanisms underlying dyspnea, there has been limited progress in treatment strategies for dyspnea. Supplemental O_2 should be administered if the resting O_2 saturation is ≤88% or if the patient's saturation drops to these levels with activity or sleep. In particular, for patients with COPD, supplemental oxygen for those with hypoxemia has been shown to improve mortality, and pulmonary rehabilitation programs (including some home and community-based exercise programs such as yoga and Tai Chi) have demonstrated positive effects on dyspnea, exercise capacity, and rates of hospitalization. More recent studies have suggested that supervised exercise programs similarly improved outcomes in post-COVID conditions. Opioids have been shown in some studies to reduce symptoms of dyspnea, largely through reducing air hunger, thus likely suppressing respiratory drive and influencing cortical activity. However, a recent study did not support the benefit of two different doses of daily low-dose extended-release morphine in COPD, for which reason opioids should be considered for each patient individually based on the risk-benefit profile in regard to respiratory depression. Studies of anxiolytics for dyspnea have not demonstrated consistent benefit.

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40 Cough

Bruce D. Levy

Cough performs an essential protective function for human airways and lungs. Without an effective cough reflex, we are at risk for retained airway secretions and aspirated material predisposing to infection, atelectasis, and respiratory compromise. At the other extreme, excessive coughing can be exhausting; can be complicated by emesis, syncope, muscular pain, or rib fractures; can aggravate low back pain, abdominal or inguinal hernias, and urinary incontinence; and can be a major impediment to social interactions. Cough is often a clue to the presence of respiratory disease. In many instances, cough is an expected and accepted manifestation of disease, as in acute respiratory tract infection; however, persistent cough in the absence of other respiratory symptoms commonly causes patients to seek medical attention.

COUGH MECHANISM

Both chemical (e.g., capsaicin) and mechanical (e.g., mucus, particulates in air pollution) stimuli can initiate a cough. Cationic channels (e.g., transient receptor potential channels) and adenosine triphosphate–activated ion channels (P2X3) function as sensory neuronal receptors, with signals transmitted centrally via Aδ (mechanosensory) and C fibers (chemosensory). Afferent nerve endings richly innervate the pharynx, larynx, and airways to the level of the terminal bronchioles and extend into the lung parenchyma. They are also located in the external auditory canal (the auricular branch of the vagus nerve, or Arnold's nerve) and in the esophagus. Sensory signals travel via the vagus and superior laryngeal nerves to a region of the brainstem in the nucleus tractus solitarius. Integrated neural networks process this input into a conscious sensation referred to as the "urge to cough." The efferent limb of the cough reflex involves a highly orchestrated series of involuntary muscular actions, with the potential for input from cortical pathways as well, making possible voluntary cough. The vocal cords adduct, leading to transient upper-airway occlusion. Expiratory muscles contract, generating positive intrathoracic pressures as high as 300 mmHg. With sudden release of the laryngeal contraction, rapid expiratory flows are generated, exceeding the normal "envelope" of maximal expiratory flow seen on the flow-volume curve (Fig. 40-1). Bronchial smooth-muscle contraction together with dynamic compression of airways narrow airway lumens and maximize the velocity of exhalation. The kinetic energy available to dislodge mucus from the inside of airway walls is directly proportional to the square of the velocity of expiratory airflow. A deep breath preceding a cough optimizes the function of the expiratory muscles; a series of repetitive coughs at successively lower lung volumes sweeps the point of maximal expiratory velocity progressively further into the lung periphery.

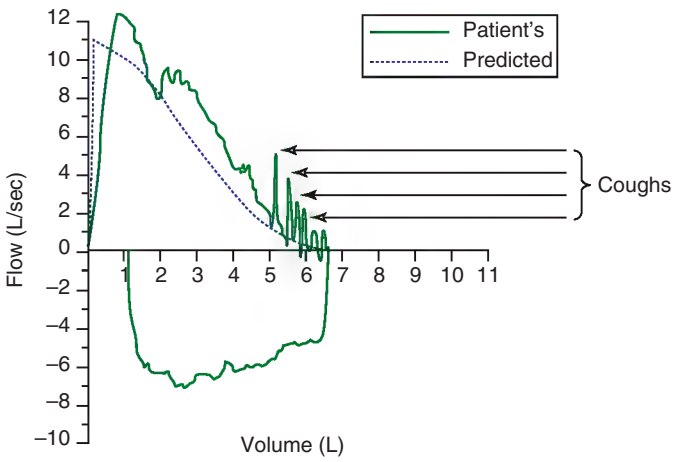

FIGURE 40-1 Flow-volume curve shows spikes of high expiratory flow achieved with cough.

TABLE 40-1 Causes of Impaired Cough and Airway Clearance

Respiratory muscle weakness
Chest wall or abdominal pain
Chest wall deformity (e.g., severe kyphoscoliosis)
Impaired glottic closure or tracheostomy
Central respiratory depression (e.g., anesthesia, sedation, or neurologic disease)
Abnormal airway secretions
Ciliary dysfunction
Tracheobronchomalacia
Bronchiectasis
Tracheal or bronchial stenoses

IMPAIRED COUGH

Weak or ineffective cough compromises the ability to clear lower respiratory tract secretions, predisposing to more serious infections and their sequelae. Weakness or paralysis of the expiratory (abdominal and intercostal) muscles and pain in the chest wall or abdomen are among the most common causes of impaired cough (Table 40-1). Cough strength is generally assessed qualitatively; peak expiratory flow or maximal expiratory pressure at the mouth can be used as a surrogate marker for cough strength. A variety of assistive devices and techniques have been developed to improve cough efficacy, running the gamut from simple (splinting of the abdominal muscles with a tightly held pillow to reduce postoperative pain while coughing) to complex (a mechanical cough-assist device supplied via face mask or tracheal tube that applies a cycle of positive pressure followed rapidly by negative pressure). Cough may fail to clear secretions completely despite a preserved ability to generate normal expiratory velocities; such failure may be due to abnormal airway secretions (e.g., abnormally viscous secretions of cystic fibrosis), ciliary dysfunction (e.g., primary ciliary dyskinesia), or structural abnormalities of the airways (e.g., tracheomalacia with excessive expiratory collapse of the trachea during cough).

SYMPTOMATIC COUGH

Cough may occur in the context of other respiratory symptoms that together point to a diagnosis; for example, cough accompanied by wheezing, shortness of breath, and chest tightness after exposure to a cat or other sources of allergens suggests asthma. At times, cough is the dominant or sole symptom of disease, and it may be of sufficient duration and severity that relief is sought. The duration of cough is a clue to its etiology, at least retrospectively. Acute cough (<3 weeks) is most commonly due to a respiratory tract infection, aspiration, or inhalation of noxious chemicals or smoke. Subacute cough (3–8 weeks in duration) is a common residuum of tracheobronchitis, as in pertussis or "postviral tussive syndrome." Chronic cough (>8 weeks) may be caused by a wide variety of cardiopulmonary diseases, including those of inflammatory, infectious, neoplastic, and cardiovascular etiologies. When initial assessment with chest examination and radiography is normal, cough-variant asthma, gastroesophageal reflux, rhinosinusitis with excessive nasopharyngeal drainage, and medications (angiotensin-converting enzyme [ACE] inhibitors) are the most common identifiable causes of chronic cough. In a long-time cigarette smoker, an early-morning, productive cough suggests chronic bronchitis. A dry, irritative cough that lingers for >2 months following one or more respiratory tract infections ("postbronchitic cough") is a very common cause of chronic cough, especially in the winter months. Chronic cough in the absence of identifiable etiology has been recognized with increasing frequency, is thought to be due to exaggerated neurologic signaling via sensory cough-reflex pathways, and is referred to as "chronic cough hypersensitivity syndrome."

ASSESSMENT OF CHRONIC COUGH

Except for our ability to detect the sound of excess airway secretions, details as to the resonance of the cough, its time of occurrence during the day, and the pattern of coughing (e.g., occurring in paroxysms) infrequently provide useful etiologic clues. Regardless of cause, cough often worsens upon first lying down at night, with talking, or with the